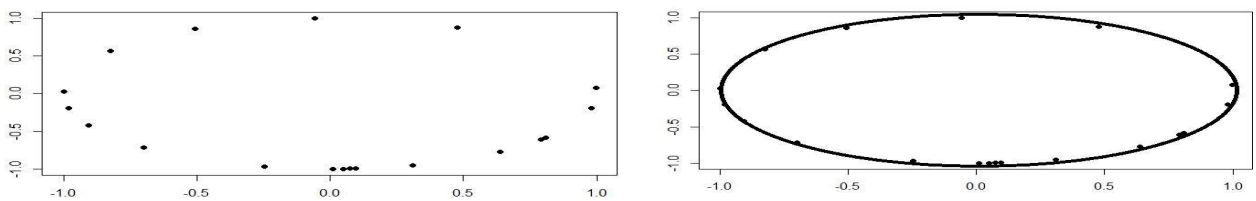

(a) observed data (b) continuous space underlying the data

Figure 1: Plots of observed data and the underlying continuous space.

$\mathbb{X}$, one has to choose a threshold, $\epsilon \geq 0$, to decide which points are close or similar to each other. For instance, we consider two points x_i and x_j in $\mathbb{X}$ to be connected, or similar, in the sense that $\delta(x_i, x_j) \leq \epsilon$. Thus, given a point cloud $(\mathbb{X}, \delta)$ and a threshold ϵ, a natural approximation of the underlying topological space is the cover of $\mathbb{X}$, *i.e*, the union of balls $\bigcup_{x \in \mathbb{X}} B_\epsilon(x) \subset \mathbb{R}^n$, where for any $x \in \mathbb{X}$, $B_\epsilon(x) = \{v \in \mathbb{R}^n : \delta(v, x) \leq \epsilon\}$. However, union of balls $\bigcup_{x \in \mathbb{X}} B_\epsilon(x)$ is not useful since it is not algorithmically tractable. Therefore, one can construct an abstract simplicial complex which is homologically equivalent to $\bigcup_{x \in \mathbb{X}} B_\epsilon(x)$. In algebraic topology, a well known theorem, **The Nerve theorem** (See: Chazal und Michel (2021)), guarantees that homology of $\bigcup_{x \in \mathbb{X}} B_\epsilon(x)$ is same as homology of the Čech complex which is defined as follows:

Definition 2.12 ***The Čech complex:*** *Suppose that we are given a point cloud $(\mathbb{X}, \delta)$, where $\mathbb{X} = \{x_1, x_2, \ldots, x_n\}$, is a set consist of observed sample points and a threshold $\epsilon \geq 0$, then one can construct an abstract simplicial complex, called Čech complex, denoted as $\mathcal{C}(\mathbb{X}, \epsilon)$, in the following way:*

- *Elements in $\mathbb{X}$ belongs to $\mathcal{C}(\mathbb{X}, \epsilon)$, that is, $\{x_i\} \in \mathcal{C}(\mathbb{X}, \epsilon)$, for all $i = 1, 2 \ldots n$.*

- *A k-simplex* $[x_{i_0}, x_{i_1} \dots x_{i_k}] \in \mathcal{C}(\mathbb{X}, \epsilon)$ *if* $\bigcap_{j=0}^{k} B_\epsilon(x_{i_j}) \neq \phi$, *where* $B_\epsilon(x)$ *is ball of radius* ϵ *around the point* x.

The Čech complex provides a procedure to construct an abstract simplicial complex from the data but its construction is not computationally tractable. Therefore, one usually constructs the following abstract simplicial complex which is computationally feasible.

Definition 2.13 ***The Vietoris-Rips complex:*** *Suppose that we are given a point cloud* $(\mathbb{X}, \delta)$, *where* $\mathbb{X} = \{x_1, x_2, \dots, x_n\}$, *is a set consisting of observed sample points and a threshold* $\epsilon \geq 0$, *then one can construct an abstract simplicial complex, called Vietoris-Rips complex or Rips complex, denoted as* $\mathcal{R}(\mathbb{X}, \epsilon)$, *in the following way:*

- *Elements in* $\mathbb{X}$ *belongs to* $\mathcal{R}(\mathbb{X}, \epsilon)$, *that is,* $\{x_i\} \in \mathcal{R}(\mathbb{X}, \epsilon)$, *for all* $i = 1, 2 \dots n$.
- *A k-simplex* $[x_{i_0}, x_{i_1} \dots x_{i_k}] \in \mathcal{R}(\mathbb{X}, \epsilon)$ *if* $B_\epsilon(x_{i_j}) \bigcap B_\epsilon(x_{i_m}) \neq \phi$ *for all* $0 \leq j < m \leq k$

The usefulness of the Rips complex is justified by the following nested relationship with the Čech complex which ensures that the Rips complex is an appropriate approximation of the Čech complex to estimate the homology of $\mathcal{X}$:

$$\mathcal{C}(\mathbb{X}, \epsilon) \subseteq \mathcal{R}(\mathbb{X}, 2\epsilon) \subseteq \mathcal{C}(\mathbb{X}, 2\epsilon).$$

2.2 Simplicial homology

Here, we shall briefly define the notion of homology for simplicial complexes. Homology counts the number of holes in a topological space. We will require the following notions to define the holes in a topological space.

Definition 2.14 ***Orientation of a simplex:*** *Given a finite simplicial complex* K, *let* σ *be a p-simplex in* K *with the vertex set* $\{x_1, \dots x_p\}$, *where* $p > 0$ *be an integer. Define two orderings of its vertex set to be equivalent if they differ from one another by an even permutation, and*

thus, the ordering of vertices falls into two equivalence classes. Each of these equivalence classes is called an ***orientation*** *of σ. We denote an oriented k-simplex with the vertex set $\{x_1, \ldots x_p\}$ as $[x_1, \ldots x_p]$.*

Definition 2.15 ***Simplicial chains:*** *Let K be a finite simplicial complex then, for any integer $p \geq 0$, a p-chain on K is defined to be formal linear combinations of p-simplices in K, that is any p-chain can be written as $\sum_{i=1}^{m} \alpha_i \sigma_i$, where σ_i is a p-simplex in K, m is the number of p-simplices in K and α_i's are integers from some field $\mathbb{F}$. An empty chain is denoted as 0 and defined as $0 = \sum_{i=1}^{m} 0.\sigma_i$.*

In TDA, one usually takes coefficients α_i's from the field $\mathbb{Z}/2\mathbb{Z} := \mathbb{Z}_2$ for computational simplicity. Here $\mathbb{Z}_2$ is a field of elements α modulo 2, where $\alpha \in \mathbb{Z}$, $\mathbb{Z}$ is the set of integers. We denote $C_p(K, \mathbb{F})$ to be the space of p-chains over the field $\mathbb{F}$. In this article, we take $\mathbb{F}$ to be $\mathbb{Z}_2$. Thus, if m denotes the number of p-simplices in K and σ_i denotes a p-simplex in K, $i = 1, \ldots, m$, then space of p-chains is defined as follows:

$$C_p(K, \mathbb{F}) := \left\{ \gamma = \sum_{i=1}^{m} \alpha_i \sigma_i : \alpha_i \in \mathbb{Z}_2 \right\}.$$

It can be verified that $C_p(K, \mathbb{F})$ is a vector space over the field $\mathbb{F}$.

Definition 2.16 ***Boundary homomorphism:*** *Given a simplicial complex K, the pth boundary homomorphism is a linear transformation from $C_p(K, \mathbb{F})$ to $C_{p-1}(K, \mathbb{F})$, also referred as boundary operator. We denote the boundary of a p-simplex σ by $\partial_p(\sigma)$.*

Let $\sigma = [x_1, x_2, \ldots, x_p]$ be an oriented p-simplex with $p > 0$, we define

$$\partial_p(\sigma) = \partial_p[x_1, x_2, \ldots, x_p] = \sum_{i=0}^{p} (-1)^i [x_1, \ldots, \hat{x}_i, \ldots, x_p],$$

where $[x_1, \ldots, \hat{x}_i, \ldots, x_p]$ is $(p-1)$- face of σ obtained by deleting x_i from $[x_1, \ldots, x_p]$. Here the symbol $\hat{x}_i$ means that we obtain $(p-1)$- face of σ by deleting the vertex x_i from $[x_1, \ldots, x_p]$.

Note that for $\mathbb{F} = \mathbb{Z}_2$, boundary of a p-simplex is defined as follows: